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qwen-qlora-train

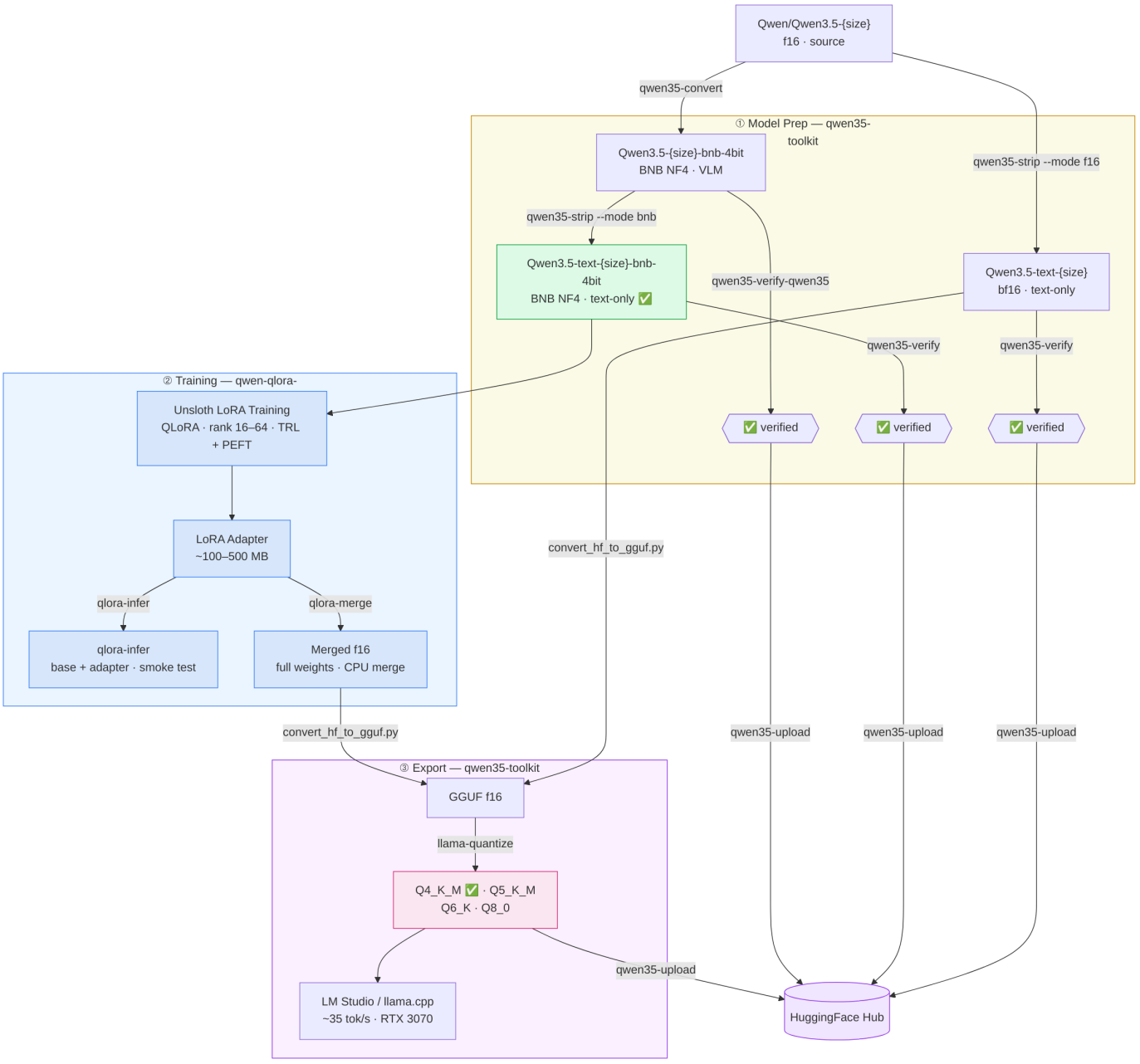
Agent-style SFT / QLoRA training pipeline for **Qwen3** and **Qwen3.5** models on limited VRAM (RTX 3070 8 GB).

Part of a two-repo ecosystem:

Repo	Purpose
qwen35-toolkit	Model prep — BNB quantization, visual tower strip, verify, upload
qwen-qlora-train (this repo)	LoRA training, adapter inference, CPU merge

---

Pipeline



Project layout

```
qwen-qlora-train/
├── pyproject.toml
├── configs/
│   ├── qwen3/
│   │   ├── 1.7b.yaml          # sanity check
│   │   ├── 4b.yaml
│   │   └── 8b.yaml            # largest validated on RTX 3070 8 GB
│   └── qwen35/
│       ├── 0.8b.yaml          # default smoke-test
│       ├── 2b.yaml
│       ├── 4b.yaml
│       └── 9b.yaml            # reference only — not a training target on 8
GB VRAM
└── qwen_qlora_train/
```

```
    ├── train.py              # entry point — orchestrates training, saves
LoRA adapter
    ├── infer.py              # entry point — base + adapter inference,
interactive chat
    ├── config.py             # TrainConfig dataclass, YAML loading,
validation
    ├── model_utils.py        # model + tokenizer loading, LoRA setup, GPU
diagnostics
    ├── data_pipeline.py      # structured truncation, char-mask,
tokenization, stats
    ├── dataset_parsers.py    # row canonicalization (reasoning fields,
tools, schemas)
    └── merge_cpu.py          # merge base + adapter on CPU into fp16/bf16
weights
```

Setup

Prerequisites

- Arch Linux (or any Linux with NVIDIA driver)
- Python 3.11
- CUDA via driver (`nvidia-smi` works → CUDA is fine)

```
# I'm using Arch btw
yay -S python311
python3.11 -m venv venv
source venv/bin/activate
```

Install

```
# 1. Unsloth — installs torch automatically with the right CUDA wheel
pip install "unsloth[cu124-torch260] @
git+https://github.com/unslothai/unsloth.git"

# 2. qwen35-toolkit
pip install git+https://github.com/techwithsergiu/qwen35-toolkit.git

# 3a. Editable install (development / local clone)
pip install -e .
```

```
# 3b. Install directly from GitHub (no local clone needed)
pip install git+https://github.com/techwithsergiu/qwen-qlora-train.git
```

> **Why this order?** Unsloth hard-pins `torch` and `transformers`. Installing it first prevents `pip install -e .` from overriding those versions.

Monitoring

```
# GPU (VRAM, utilization)
nvtop
# or
watch -n 1 nvidia-smi

# RAM
htop
# or
free -h
```

Training

Qwen3 (validated)

Default smoke-test model: **`unsloth/Qwen3-1.7B-bnb-4bit`**

```
# 1. Check token length distribution
qlora-train --config configs/qwen3/1.7b.yaml --stats-only

# 2. Inspect 2 processed samples (what the model actually trains on)
qlora-train --config configs/qwen3/1.7b.yaml --debug-samples 2

# 3. Train
qlora-train --config configs/qwen3/1.7b.yaml
```

Outputs:

- LoRA adapter: `adapters/<run_name>/`
- Checkpoints: `outputs/<run_name>/`

Config	Model	max_length
`configs/qwen3/1.7b.yaml`	`unsloth/Qwen3-1.7B-bnb-4bit`	5120
`configs/qwen3/4b.yaml`	`unsloth/Qwen3-4B-bnb-4bit`	5120
`configs/qwen3/8b.yaml`	`unsloth/Qwen3-8B-bnb-4bit`	1024

> **Hardware note:** 8B was previously validated on RTX 3070 8 GB with `max_length: 1024`. Newer library versions (unsloth 2026.3.4+) allocate memory more conservatively, leaving ~500 MB headroom that is no longer available for training — the run now OOMs before the first step. The config is kept for when this is resolved. See OOM troubleshooting.

Qwen3.5 (configs ready, end-to-end not yet validated)

Prepare base models with `qwen35-toolkit` before training:

```
qwen35-strip --model unsloth/Qwen3.5-0.8B --output ./Qwen3.5-text-0.8B --
mode f16
qwen35-strip --model ./Qwen3.5-text-0.8B --output ./Qwen3.5-text-0.8B-bnb-
4bit --mode bnb
qwen35-verify --model ./Qwen3.5-text-0.8B-bnb-4bit
```

> `chat_template: qwen3` — Qwen3.5 uses the same `<|im_start|>/<|im_end|>` format as Qwen3. Unsloth has no separate key for Qwen3.5.

Default smoke-test model: **`techwithsergiu/Qwen3.5-text-0.8B-bnb-4bit`**

```
qlora-train --config configs/qwen35/0.8b.yaml --stats-only
qlora-train --config configs/qwen35/0.8b.yaml --debug-samples 2
qlora-train --config configs/qwen35/0.8b.yaml
```

Config	Model	max_length
`configs/qwen35/0.8b.yaml`	`techwithsergiu/Qwen3.5-text-0.8B-bnb-4bit`	5120
`configs/qwen35/2b.yaml`	`techwithsergiu/Qwen3.5-text-2B-bnb-4bit`	5120
`configs/qwen35/4b.yaml`	`techwithsergiu/Qwen3.5-text-4B-bnb-4bit`	5120
`configs/qwen35/9b.yaml`	`techwithsergiu/Qwen3.5-text-9B-bnb-4bit`	1024

> **Hardware note:** 9B is **not a realistic training target on RTX 3070 8 GB**. Even with 4-bit quantization and `max_length: 1024`, the model weights alone (~8 GB) leave almost no headroom for activations, KV cache, and optimizer state. The config exists as a reference. Use 0.8B or 2B for actual training runs.

Dataset handling and templating

How it works

The pipeline never builds prompts manually. Instead it works in three layers:

```
HF dataset row
      |
      ▼  dataset_parsers.py — canonicalize_row()
      |  • detect schema (messages / prompt_response / auto)
      |  • normalize reasoning fields → reasoning_content
      |  • extract <think>...</think> from assistant content
      |  • apply think_mode / think_max_tokens policy
      |  • detect tools column → extras{"tools": ...}
      |
      ▼  data_pipeline.py — structured_truncate_messages()
      |  • prune middle history turns to fit max_length
      |  • trim reasoning_content in kept turns
      |  • fallback: token-level left-truncation
      |
      ▼  data_pipeline.py — build_text_and_masks()
      |  • tokenizer.apply_chat_template(messages, tools=tools)
      |  • char-level mask: 1 = loss, 0 = masked
      |  • assistant_roles masking
      |  • think_loss sub-span masking
```

```
        │
        ▼  data_pipeline.py — tokenize_with_char_mask()
           • tokenize + project char mask → token labels
           • returns {input_ids, attention_mask, labels}
```

The chat template (Jinja2) lives in the tokenizer — we just call it. `get_chat_template(tokenizer, chat_template="qwen3")` patches it once at load time so the correct template is always used regardless of what the Hub model ships with.

Supported dataset schemas

`messages` (default — OpenAI API format)

The most common format. Detected when the row has a list under `messages_field`.

```
{
  "messages": [
    {"role": "system",    "content": "You are a helpful assistant."},
    {"role": "user",      "content": "What is 2+2?"},
    {"role": "assistant", "content": "4"}
  ]
}
```

Config:

```
dataset_schema: auto      # or "messages" explicitly
messages_field: messages  # column name
```

`prompt_response`

Detected when the row has `prompt` + one of `response` / `completion` / `answer`. Converted to a two-message conversation internally.

```
{"prompt": "What is 2+2?", "response": "4"}
```

Config:

```
dataset_schema: auto   # auto-detected, no extra config needed
```

Reasoning / thinking fields

Qwen3 stores reasoning inside assistant messages as `reasoning_content`. Many datasets use different field names — all of these are recognized and normalized to `reasoning_content` before the template sees them:

```
reasoning_keys: ["reasoning", "reasoning_content", "thinking", "reason"]
```

Also handles `<think>...</think>` inline inside `assistant.content`:

```
{"role": "assistant", "content": "<think>let me think...</think>
4"}
```

→ extracted into `reasoning_content`, content becomes `"4"`.

Tools support

Tool schemas are auto-detected from common column names and passed directly to `apply_chat_template(messages, tools=tools)` — the Qwen template injects them into the system prompt automatically.

Detected columns (first match wins): `tools`, `tool_schemas`, `functions`, `function_schemas`.

Tools can be stored as a list, dict, or JSON string — all normalized.

> ⚠️ Tool schemas can add hundreds or thousands of tokens per sample. Always run `--stats-only` first when training on tool datasets.

Assistant-only loss masking

```
assistant_roles: ["assistant"]
```

Loss is computed **only on spans produced by roles in `assistant_roles`**. System / user / tool messages are masked with `-100`.

Masking is done in character space before tokenizing — no tokenizer-specific heuristics, works correctly with any subword vocabulary.

think_loss — sub-span control inside assistant turns

```
think_loss: all             # all | answer_only | answer_plus_think
```

- `all` — train on full assistant span (think + answer)
- `answer_only` — mask everything up to and including `</think>`, train only on the final answer
- `answer_plus_think` — mask the literal `<think>`/`</think>` tags, train on content + answer

Truncation strategy

```
truncate_side: left   # always recommended for chat
max_length: 4096
```

Naive left-truncation (keep last N tokens) destroys the system prompt and tool schemas. The pipeline uses **structured truncation** instead:

1. Drop oldest middle turns first (keep system + last user + last assistant)
2. Trim `reasoning_content` in kept turns
3. Fallback: token-level left-truncation only if still over `max_length`

Dataset diagnostics

```
qlora-train --config configs/qwen35/0.8b.yaml --stats-only
```

Output:

```
[Token length stats]  n=247  max_length=5120

  Distribution
    p25 :    360
    p50 :    567
    p75 :   1741
    p90 :   3608
    p95 :   4404
    p99 :   5120  ← above max_length
    max :   5120  ← above max_length  (dataset max: 8327)

  Window utilisation (tokens / max_length)
    < 25% :   180 samples  ( 72.9%)   [▓▓▓
```

think_mode

```
think_mode: keep   # keep | drop
```

- `keep` — preserve reasoning (recommended)
- `drop` — remove reasoning entirely (pure answer-style)

think_loss

```
think_loss: all   # all | answer_only | answer_plus_think
```

- `all` — train on full assistant span (think + answer)
- `answer_only` — mask everything up to `</think>`, train only on final answer
- `answer_plus_think` — mask only the `<think>`/`</think>` tags

Recommended patterns:

```
# Agent / tool training — stable answers
think_loss: answer_only
think_mode: keep

# Reasoning dataset — train full chain of thought
think_loss: all
think_mode: keep
```

Adapter inference

Test a LoRA adapter directly on the base model — **no CPU merge needed**. This is the primary way to verify an adapter immediately after training.

```
# Predefined smoke tests (4 prompts covering short answer, code, stop,
think)
qlora-infer \
  --model   unsloth/Qwen3-1.7B-bnb-4bit \
  --adapter adapters/qwen3-1.7b-sanity
```

```
# Single prompt
qlora-infer \
  --model   unsloth/Qwen3-1.7B-bnb-4bit \
  --adapter adapters/qwen3-1.7b-sanity \
  --prompt  "Explain LoRA in two sentences."

# Interactive chat loop (maintains full conversation history)
qlora-infer \
  --model   unsloth/Qwen3-1.7B-bnb-4bit \
  --adapter adapters/qwen3-1.7b-sanity \
  --interactive

# Disable thinking mode
qlora-infer \
  --model   unsloth/Qwen3-1.7B-bnb-4bit \
  --adapter adapters/qwen3-1.7b-sanity \
  --no-thinking
```

Works with any model/adapter combo — `--backend auto` detects 4-bit vs fp16 from the model id.

Flag	Default	Description
`--model`	—	HF repo id or local path (**required**)
`--adapter`	`null`	LoRA adapter directory
`--backend`	`auto`	`auto` detects from model id · `unsloth` · `transformers`
`--dtype`	`f16`	`f16` or `bf16`
`--chat-template`	`qwen3`	Unsloth template key (unsloth path only)
`--max-new`	`1024`	Max tokens to generate
`--temp`	`0.7`	Sampling temperature
`--top-p`	`0.9`	Top-p sampling
`--prompt`	`null`	Single user prompt — skips predefined tests
`--interactive`	`false`	Interactive chat loop with conversation history
`--no-thinking`	`false`	Set `enable_thinking=False` in chat template

CPU merge

Needed only if you want a standalone merged model (for GGUF conversion or publishing). If you just want to test the adapter, use `qlora-infer` instead — no merge needed.

```
# Qwen3.5 0.8B
qlora-merge \
  --base    unsloth/Qwen3.5-0.8B \
  --adapter adapters/qwen35-text-0.8b-sanity \
  --output  merged/qwen35-text-0.8b-merged-f16 \
  --dtype   f16

# Qwen3 1.7B
qlora-merge \
  --base    unsloth/Qwen3-1.7B \
  --adapter adapters/qwen3-1.7b-sanity \
  --output  merged/qwen3-1.7b-merged-f16 \
  --dtype   f16
```

Hardware: ~10–20 GB RAM depending on model size, no VRAM needed.

Post-merge workflow

After `qlora-merge` you have a standalone fp16 model. The next steps — GGUF conversion, quantization, and Hub upload — are handled by **qwen35-toolkit**.

The full pipeline is documented there. Summary of what happens:

Step	Tool	What it produces
Convert to GGUF	`llama.cpp/convert_hf_to_gguf.py`	`model-F16.gguf`
Quantize	`llama-quantize`	`Q4_K_M` · `Q5_K_M` · `Q6_K` · `Q8_0`
Upload	`qwen35-upload`	HuggingFace Hub repo

> See **qwen35-toolkit → Usage examples** for the exact commands — Steps 3 and 4 in that section cover GGUF conversion and upload.

Config reference

All fields with their defaults. Any field omitted from YAML uses the default.

Identity

Field	Default	Description
`run_name`	`"run"`	Subdirectory name under `output_dir/` and `adapter_base_dir/`
`output_dir`	`"outputs"`	Root dir for trainer checkpoints and logs
`adapter_base_dir`	`"adapters"`	Root dir for saved LoRA adapter
`hf_token`	`null`	HF access token — prefer `HF_TOKEN` env var

Model

Field	Default	Description
`model_name`	`"unsloth/Qwen3-4B-bnb-4bit"`	HF repo id or local path
`chat_template`	`"qwen3"`	Unsloth template key — `"qwen3"` works for both Qwen3 and Qwen3.5

Dataset

Field	Default	Description
`dataset_id`	`""`	HF dataset id or local path (**required**)
`dataset_split`	`"train"`	Split passed to `load_dataset`
`messages_field`	`"messages"`	Column name containing the conversation list

Field	Default	Description
`dataset_schema`	`"auto"`	`"auto"` / `"messages"` / `"prompt_response"`

Reasoning / thinking

Field	Default	Description
`reasoning_field`	`"reasoning_content"`	Canonical field the Qwen template reads for `<think>` content
`reasoning_keys`	`null`	Alternative field names to normalize into `reasoning_field`. Default: `["reasoning_content", "reasoning", "thinking", "reason"]`
`extract_think_tags`	`true`	Extract inline `<think>…</think>` from `assistant.content`
`think_mode`	`"keep"`	`"keep"` preserve reasoning · `"drop"` remove entirely
`think_max_tokens`	`0`	Hard token cap on reasoning per message. `0` = no cap
`think_role`	`"think"`	Role name for datasets that store thinking as a separate message (rare)
`think_loss`	`"all"`	`"all"` · `"answer_only"` · `"answer_plus_think"` — see below

`think_loss` values:

Value	What gets gradient
`all`	Full assistant span — `<think>` content + answer
`answer_only`	Only tokens after `</think>` — recommended for agent/tool training
`answer_plus_think`	Think content + answer, but not the literal `<think>`/`</think>` tags

Sequence / truncation

Field	Default	Description
`max_seq_length`	`2048`	Model's position embedding allocation (FastLanguageModel). Set ≥ `max_length`
`max_length`	`2048`	Max tokens per training sample. **Primary VRAM lever on 8 GB**
`truncate_side`	`"left"`	Fallback token-level cut: `"left"` keeps last N tokens (recommended)

Precision / hardware

Field	Default	Description
`load_in_4bit`	`true`	BNB NF4 quantization — required for >1.7B on 8 GB VRAM
`attn_implementation`	`"sdpa"`	Attention backend — `"sdpa"` or `"flash_attention_2"`
`fp16`	`true`	Float16 training — use for Qwen3
`bf16`	`false`	BFloat16 training — use for Qwen3.5. Mutually exclusive with `fp16`

LoRA

Field	Default	Description
`lora_r`	`16`	LoRA rank. Higher = more capacity + VRAM
`lora_alpha`	`32`	Scaling factor. Convention: `2 * lora_r`
`lora_dropout`	`0.0`	Dropout on LoRA layers. Keep `0.0` with gradient checkpointing
`lora_target_modules`	`null`	Layers to apply LoRA to. `null` = all 7 projection layers
`gradient_checkpointing`	`"unsloth"`	`"unsloth"` (recommended, ~30% less VRAM) · `true` · `false`

Training

Field	Default	Description
`per_device_train_batch_size`	`1`	Keep at 1 for 8 GB VRAM
`gradient_accumulation_steps`	`8`	Effective batch = batch_size × accumulation_steps
`learning_rate`	`2e-4`	Peak LR for AdamW
`warmup_ratio`	`0.05`	Fraction of steps used for LR warmup. ⚠️ Deprecated in TRL v5.2 — will be replaced by `warmup_steps`
`max_steps`	`1000`	Total optimizer steps
`logging_steps`	`20`	Log loss every N steps
`save_steps`	`200`	Save checkpoint every N steps
`seed`	`3407`	Random seed for torch, LoRA init, and SFTConfig

Field	Default	Description
`optim`	`"adamw_8bit"`	`"adamw_8bit"` (saves ~1 GB VRAM) or `"adamw_torch"`

Loss masking

Field	Default	Description
`assistant_roles`	`null`	Roles that carry loss. `null` = `["assistant"]`. All others masked with `-100`
`drop_if_no_assistant`	`true`	Skip samples with no assistant turn — they produce zero loss and waste compute

---

OOM troubleshooting (RTX 3070 8 GB)

Apply in order — `max_length` is the strongest lever:

1. Reduce `max_length` — KV cache and activations scale with sequence length
2. Reduce `lora_r`
3. Keep `per_device_train_batch_size: 1`
4. Increase `gradient_accumulation_steps` — maintains effective batch size

If using tool-schema datasets, schemas can add thousands of tokens per sample. Always run `--stats-only` first.

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LICENSE

Third-party Licenses

This project relies on several third-party components, all using permissive licenses compatible with Apache License 2.0.

- **qwen35-toolkit** — Apache License 2.0 (© TechWithSergiu)
  github.com/techwithsergiu/qwen35-toolkit
- **Unsloth** — Apache License 2.0 (© Unsloth AI)
  github.com/unslothai/unsloth
- **PyTorch** — BSD 3-Clause License (© PyTorch Contributors)
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  github.com/yaml/pyyaml